

Customer Operations Automation

Customer Onboarding Workflow — Portfolio Demonstration

INDUSTRY	PROJECT TYPE	ROLE	STATUS
Fitness / Wellness	Automation & operations workflow demonstration	AI & Automation Engineer	Working local demonstration

01 The business problem

As a customer-facing business grows, onboarding can become increasingly dependent on manual coordination: payment, customer information, onboarding form, validation, staff assignment, customer communication and activation.

The problem is not necessarily a lack of software. It is often that multiple systems and people are involved in the same customer journey without a coordinated workflow.

02 The demonstrated solution

A working demonstration of an **automated customer onboarding workflow**. It coordinates the process from the initial customer/payment event through to customer activation.

The system maintains each customer's current state and records the operational events that occur along the way.

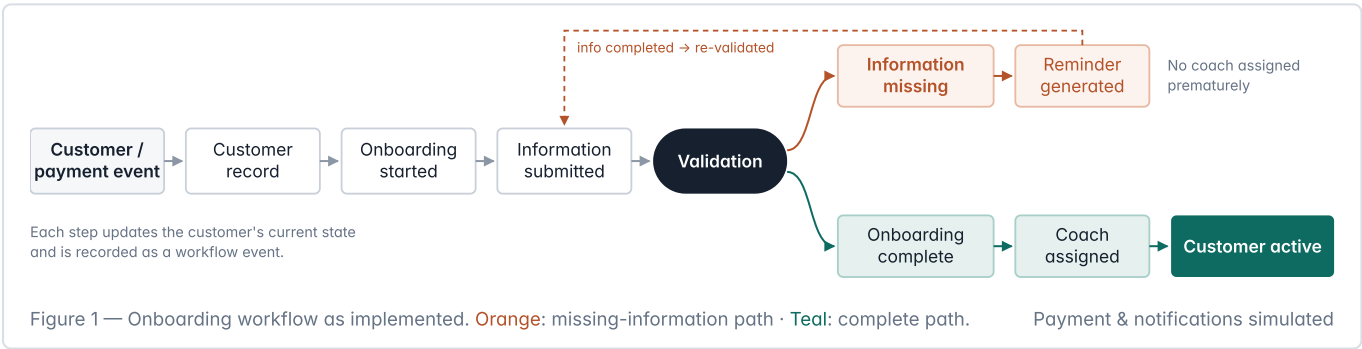
WHEN THESE STEPS ARE HANDLED MANUALLY, TEAMS REPEATEDLY CHECK:

? Who has paid?
 ? Who needs a reminder?

? Who has completed onboarding?
 ? Which staff member is handling the customer?

? What information is still missing?
 ? Has the customer actually been activated?

03 Automated workflow



04 What the demonstration does

1 · CREATE	2 · VALIDATE	3 · FOLLOW UP	4 · COMPLETE	5 · ASSIGN
Customer creation A simulated payment event creates the customer record, starts onboarding and sends a simulated onboarding-form notification.	Information validation Submitted onboarding information is checked for the required details.	Missing information Customer marked <i>Information Missing</i> , missing fields identified, reminder generated. No coach is assigned prematurely.	Onboarding completion Onboarding marked complete, coach assigned and a simulated welcome notification sent.	Coach assignment The least-loaded coach is assigned and recorded. Reassignment does not re-send the welcome message.

05 Scenario highlights — each scenario tested end-to-end in the local environment

1	Complete onboarding TESTED LOCALLY	Customer created → Onboarding form completed → Information validated → Coach assigned → Customer active Full path from payment event to an active customer, with each state change recorded.
2	Missing information TESTED LOCALLY	Incomplete info submitted → Missing fields identified → Marked pending → Reminder generated → Completed → workflow continues Incomplete onboarding is caught and followed up. Coach assignment waits until information is complete.
3	Coach reassignment TESTED LOCALLY	Completed customer → Assigned to a coach → Coach changed → Reassignment recorded → New coach notified Ownership change is tracked. The customer does not receive the initial welcome message again.

06 Operational visibility

The operations dashboard gives the team a central view of where customers are in the onboarding process. It provides visibility into:

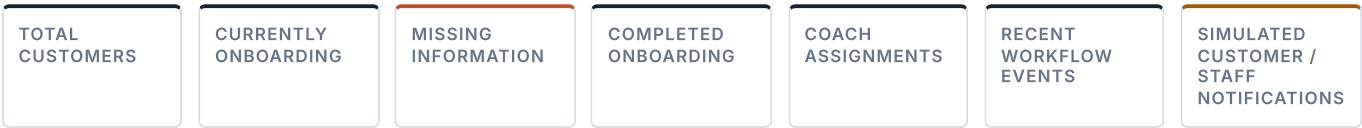


Figure 2 — Dashboard views in the demonstration (list of views, not a screenshot). Customer and staff notifications are simulated

07 Technology

n8n	Workflow orchestration
FastAPI	Backend / API and business logic
MySQL	Operational data and workflow state
SQLAlchemy	Database access
React + Vite	Operations dashboard
Docker Compose	Local environment

The implementation uses API-driven integration between the workflow orchestration layer and the application backend.

08 What is simulated

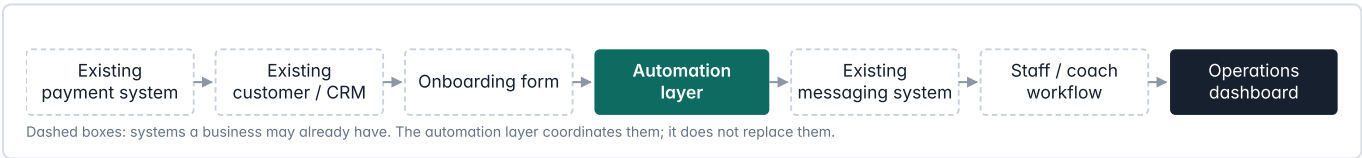
This is a portfolio demonstration, not a deployment for a real business.

- Payment events are simulated in a **Razorpay-like format**.
- WhatsApp messages are simulated and stored in a **notification log**.
- Onboarding form submission is simulated **through the dashboard**.
- Coaches are **fictional demo users**.

No real customer data or paid external services are required. The architecture is designed so these simulated components can later be replaced with a client's actual systems and APIs.

09 Real-world adaptation

The same workflow pattern can connect to a business's **existing** tools rather than replacing them. Integrations could be built around payment systems, forms, messaging platforms, CRM systems, databases or internal applications, depending on what the business already uses.



10 Business value

Reduce repetitive coordination

Predictable steps run automatically instead of staff manually initiating every action.

Reduce missed onboarding steps

Incomplete information is tracked and the appropriate follow-up is triggered.

Improve ownership

Assignments and workflow states are visible, so it is clear who handles each customer.

Improve operational visibility

The team has a central view of where every customer is in onboarding.

No specific ROI or time savings are claimed. This is a demonstration and has not yet been deployed in a real business.

11 How I approach client projects

I don't start by forcing a business into a predefined software product. The existing business process comes first, and the technology is designed around it.



12 Project status Working portfolio demonstration — tested locally

